



Batterjee Medical College
For Science and Technology

An approach to a case of Wheezing Asthma

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Learning Objectives

Knowledge:

- Define asthma as a chronic inflammatory disease characterized by bronchial hyperresponsiveness and reversible airflow obstruction, and describe its global epidemiology.
- Differentiate between allergic and nonallergic asthma, as well as clinical subtypes.
- Explain Th2-cell-mediated inflammation, Type-I hypersensitivity, mast cell degranulation, smooth muscle hypertrophy, mucosal edema, and mucus plugging.
- List core symptoms and physical signs.
- Identify standard diagnostic tests and explain the stepwise classification from mild intermittent to severe persistent asthma.
- Outline controller/reliever drug classes and management protocols for acute asthma exacerbations.

Skills:

- Interpret patient histories, physical findings, spirometry metrics peak flow readings, ABGs, and chest radiographs.
- Apply clinical parameters to classify asthma severity and stage acute exacerbations.
- Formulate evidence-based treatment regimens using stepwise management guidelines and construct acute emergency management plans.
- Differentiate asthma from other causes of acute or chronic dyspnea, particularly COPD.

Values:

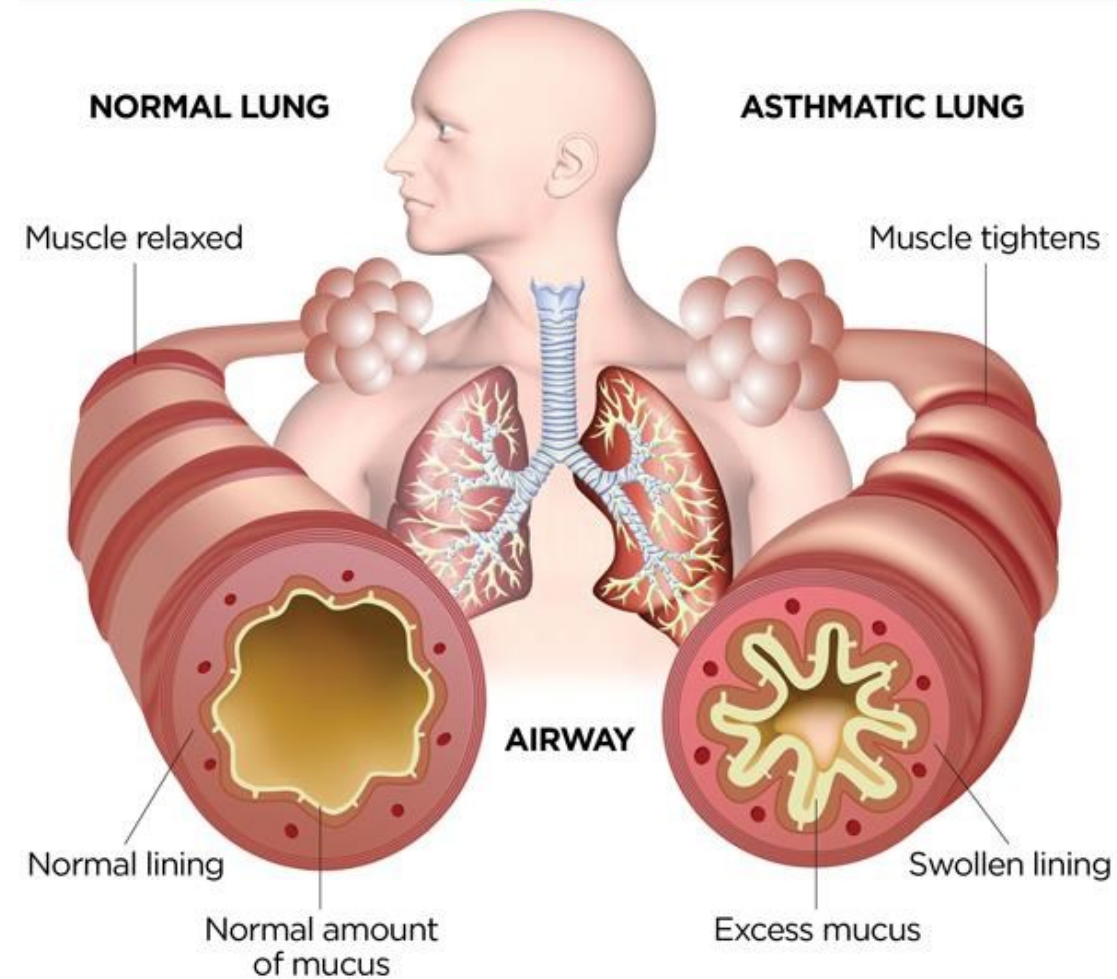
- Demonstrate a professional attitude through strict compliance with training site rules, regulations, and attendance policies.
- Value patient education regarding proper inhaler techniques, medication adherence, trigger avoidance, and self-monitoring.



- Definition
- Epidemiology
- Etiology
- Pathophysiology
- Clinical Features
- Differential Diagnosis
- Diagnostics
- Classification
- Management
- Subtypes
- Acute Asthma exacerbation

Definition

- A chronic **inflammatory** disease of the respiratory system characterized by bronchial **hyperresponsiveness**, which manifest with respiratory symptoms, and a **reversible airflow obstruction**



Epidemiology & Etiology

Types:

- Allergic asthma: typically in **childhood**
- Nonallergic asthma: typically **> 40 years**

Risk Factors:

- Family history of asthma
- Allergy
- Atopy
- Environmental & socioeconomic factors (Pollution from cooking and heating)

Etiology

CHARACTERISTIC	ALLERGIC ASTHMA	NONALLERGIC ASTHMA
Synonym	Extrinsic asthma	Intrinsic asthma
Onset	Early	Late
Worsening with allergens	+	–
Personal history of atopy	+	–
Family history of atopy	+	–
Intolerance to NSAIDs	Less frequent	More frequent
Pathophysiology	Type I hypersensitivity	Unknown
Total IgE	Elevated	Normal
Specific IgE	+	–
Eosinophils (blood/sputum)	+	+ or –
Evolution	More favorable	Less favorable

• **Nonallergic asthma: is Irritant asthma mostly with Aspirin or NSAIDs**

Pathophysiology (common)

Asthma is an inflammatory disease driven by **T-helper type 2 cells (Th2-cell)** that manifests in individuals with a **genetic predisposition**. It consists of the following **three pathophysiologic** processes:

1. Bronchial hyperresponsiveness

2. Bronchial inflammation

inhalation of antigen → activation of eosinophils → bronchial submucosal edema → bronchioles collapse

3. Endobronchial obstruction caused by:

Increased parasympathetic tone leads to

Reversible bronchospasm

Increased mucus production

Mucosal edema with hyperplasia of goblet cells

Clinical Features

HPI:

- Symptoms appear in response to a variety of stimuli (Dust)
- Usually asymptomatic between attacks
- **During attacks:**
 - Dyspnea
 - Cough (persistent dry cough, ↑ exercise, exposure to irritants and worse at night)
 - Chest tightness
 - Wheezing

Past Medical History:

- Atopy, eczema, atopic dermatitis, allergic rhinitis, or allergic conjunctivitis

Family & Social History:

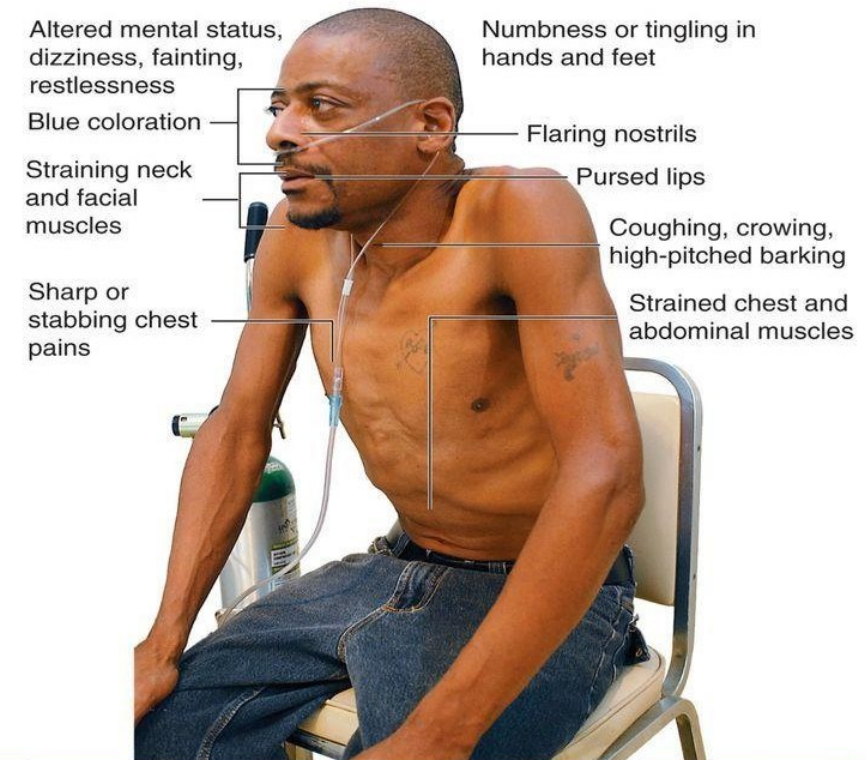
- Atopy or asthma
- mold, carpet, tobacco, pet exposure

Clinical Features

Signs:

- Wheezing
- Prolonged expiratory phase
- Silent chest in sever cases
- Hyperresonance on percussion
- Tachypnea
- Tachycardia
- Cyanosis
- Lethargy
- Pulsus paradoxus

Signs and symptoms of respiratory compromise.



Subtypes & Variants

•Allergic asthma

- Most common phenotype
- Begins with intermittent symptoms in childhood
- Triggered by allergens
- Usually associated with atopy (e.g., eczema, rhinitis)
- Responds well to ICS-containing treatment

•Nonallergic asthma

- Less common than allergic asthma
- Triggered by, e.g., viral upper respiratory tract infections, cold air, GERD
- Not associated with atopy
- Responds poorly to ICS-containing treatment

Subtypes & Variants

- **Adult-onset asthma:**

- An uncommon phenotype in which symptoms first manifest in adulthood
- Typically nonallergic
- Responds poorly to ICS-containing treatment

- **Cough variant asthma:** a type of asthma characterized by chronic dry cough without other typical symptoms of asthma

- **Aspirin-exacerbated respiratory disease**

- **Asthma-COPD overlap**

- **Occupational asthma**

Subtypes & Variants

Exercise induced Asthma

Asthma symptoms occurring within 15 minutes of initiating intense exercise and resolves within an hour

Management:

- Non-pharmacological therapy
- Inhaled beta-agonists 15–30 minutes before exercise
- If required ≥ 4 times a week --> daily ICS therapy

Diagnostics

PFT (Spirometry)

- ↓ FEV1/FVC ratio ($< 70\%$) obstructive pattern
- ↓ FEV1
- reversible obstruction ($>12\%$ ↑ in FEV1 after bronchodilator use)

Bronchoprovocation test

- ↓ FEV1 $>20\%$ with methacholine or histamine

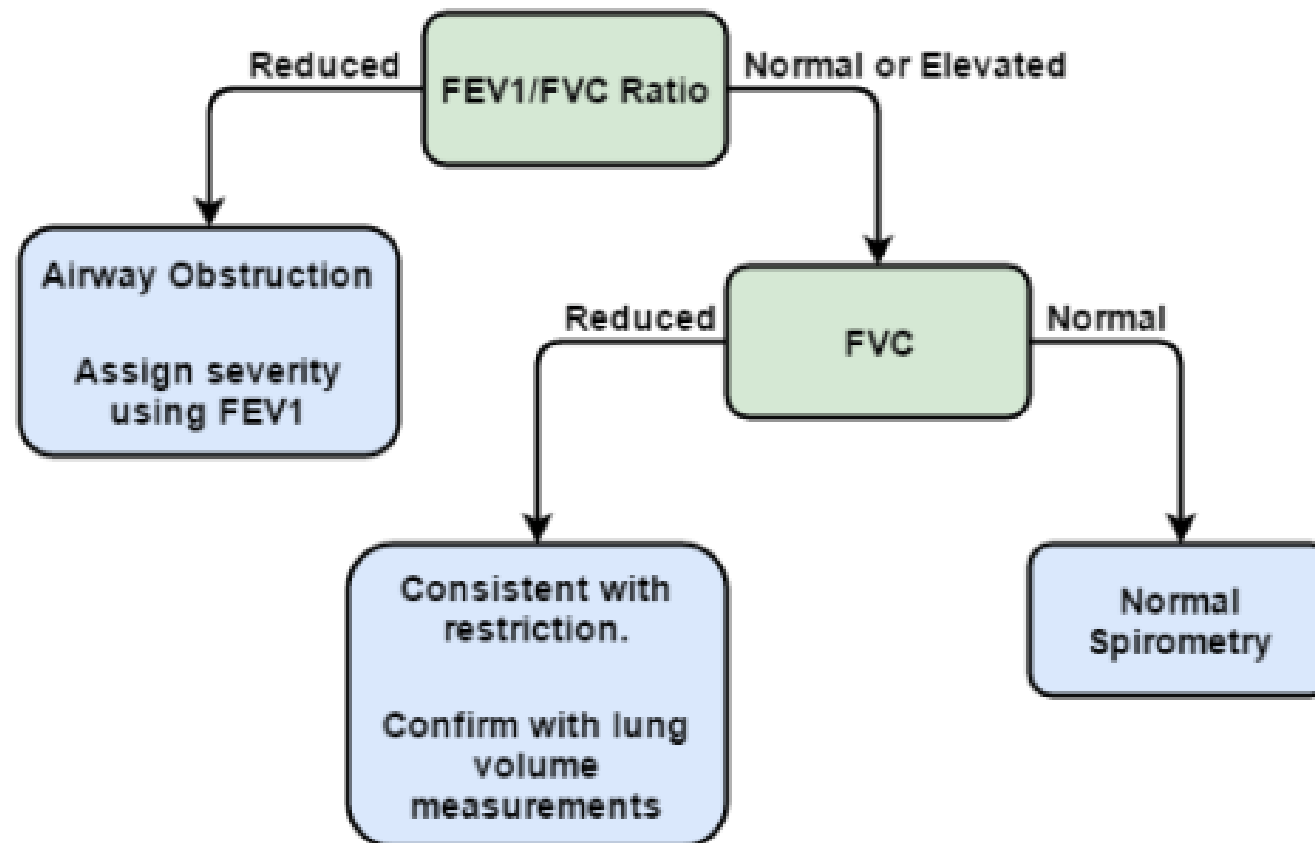
Peak expiratory flow rate (PEF)

- The expected peak expiratory flow rate will vary depending on the patient's **age**, **sex** and **height**
- A normal reading is approximately between **400 – 600** L/min.



Diagnostics

PFT (Spirometry)



Diagnostics

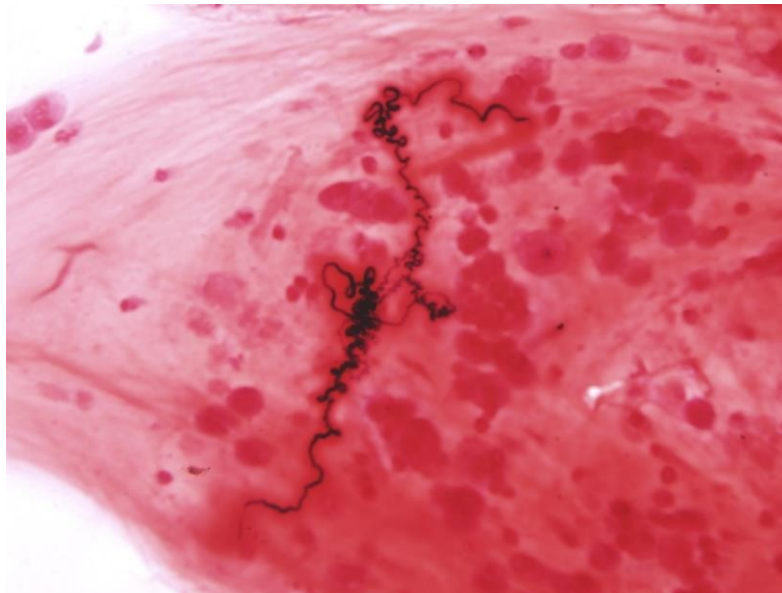
- CBC: possible **eosinophilia**
- Antibody testing: ↑
total IgE, ↑ allergen-specific IgE
- Skin allergy tests: skin prick testing or
intradermal skin testing
- ABG: BAD prognosis if hypercapnia is
present



Diagnostics

Sputum:

- Sputum may appear purulent
- Curschmann's spirals: mucus plug in sputum that is formed from shed bronchial epithelium
- Charcot–Leyden crystals: histopathologic finding in patients with eosinophilic inflammation and/or proliferation



Diagnostics

CXR

- Normal in mild to moderate cases
- Severe asthma reveals hyperinflation.
- Necessary in severe asthma to exclude other conditions

ABG

- Hypocarbia is common
- Normal/Increased PaCO_2 is a bad sign

CBC

- Leukocytosis and eosinophilia



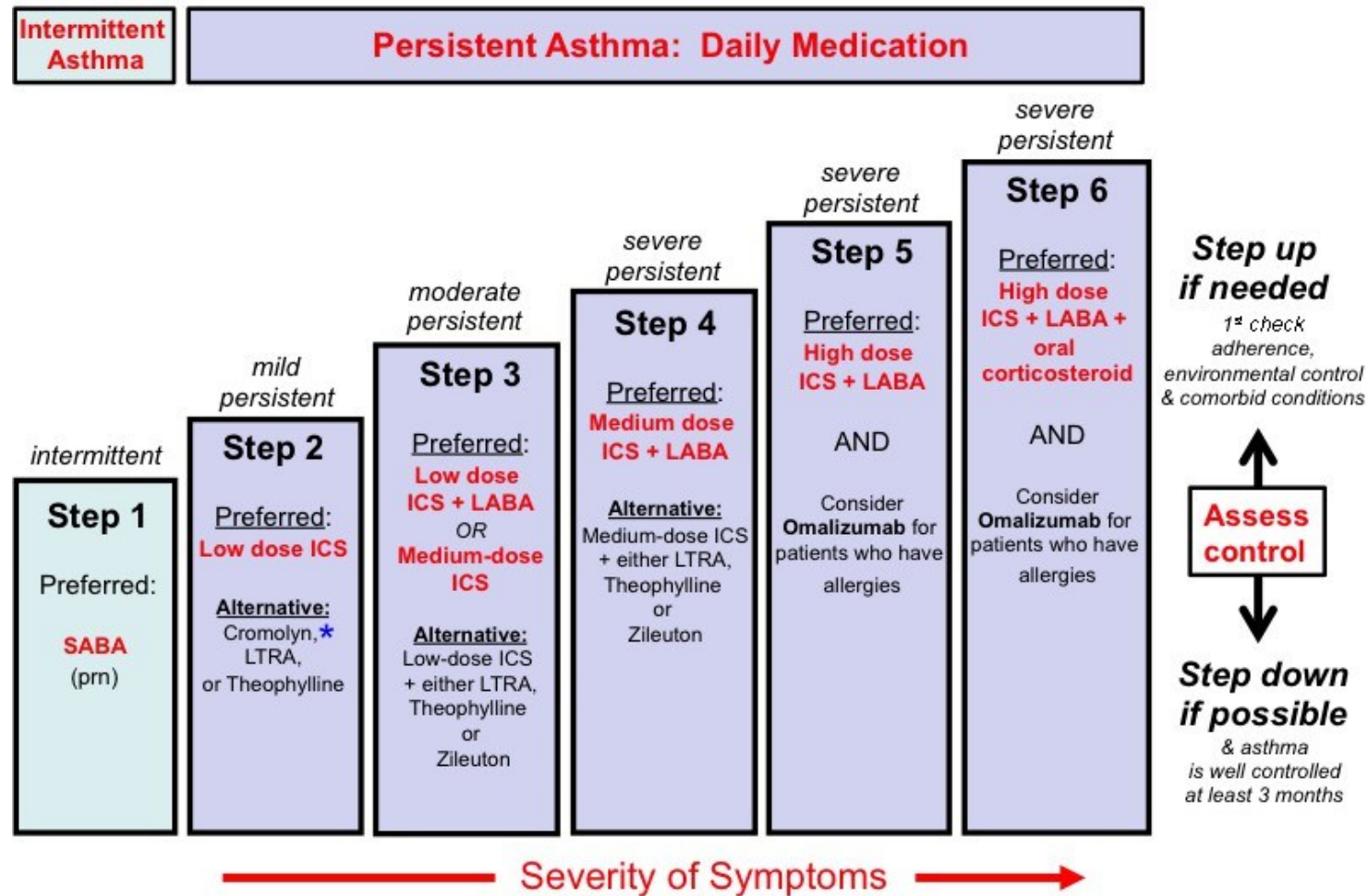
Management

Asthma Classification

Classification	Daytime Symptoms	Nighttime Symptoms	PEF or FEV1
Mild intermittent	≤ 2 per week	< 2 per month	$\geq 80\%$
Mild persistent	> 2 per week, but < 1 per day	> 2 per month	$\geq 80\%$
Moderate persistent	Daily symptoms	> 1 per week	60%-80%
Severe persistent	Continued day symptoms	Frequent night symptoms	$< 60\%$

Management

Stepwise Approach for Managing Asthma in Adults



Management

- Inhaled β_2 -agonists
- Corticosteroids (inhaled & PO)
- Montelukast (leukotriene modifiers)
- Theophylline
- anti-IgE monoclonal antibody (omalizumab)

	LABA	LAMA	Inhaled steroids
12 hours	Salmeterol	Acclidinium	Beclomethasone
	Formoterol	Darotropium	Budesonide
24 hours			Fluticasone propionate
	Indacaterol	Tiotropium	Ciclesonide
	Vilanterol	Glycopyrronium	Mometasone
	Olodaterol	Umeclidinium	Fluticasone furoate
	Abediterol		
	Carmoterol		

Abbreviations: COPD, chronic obstructive pulmonary disease; LABA, long-acting beta agonist; LAMA, long-acting muscarinic antagonist.

Acute Asthma Exacerbation

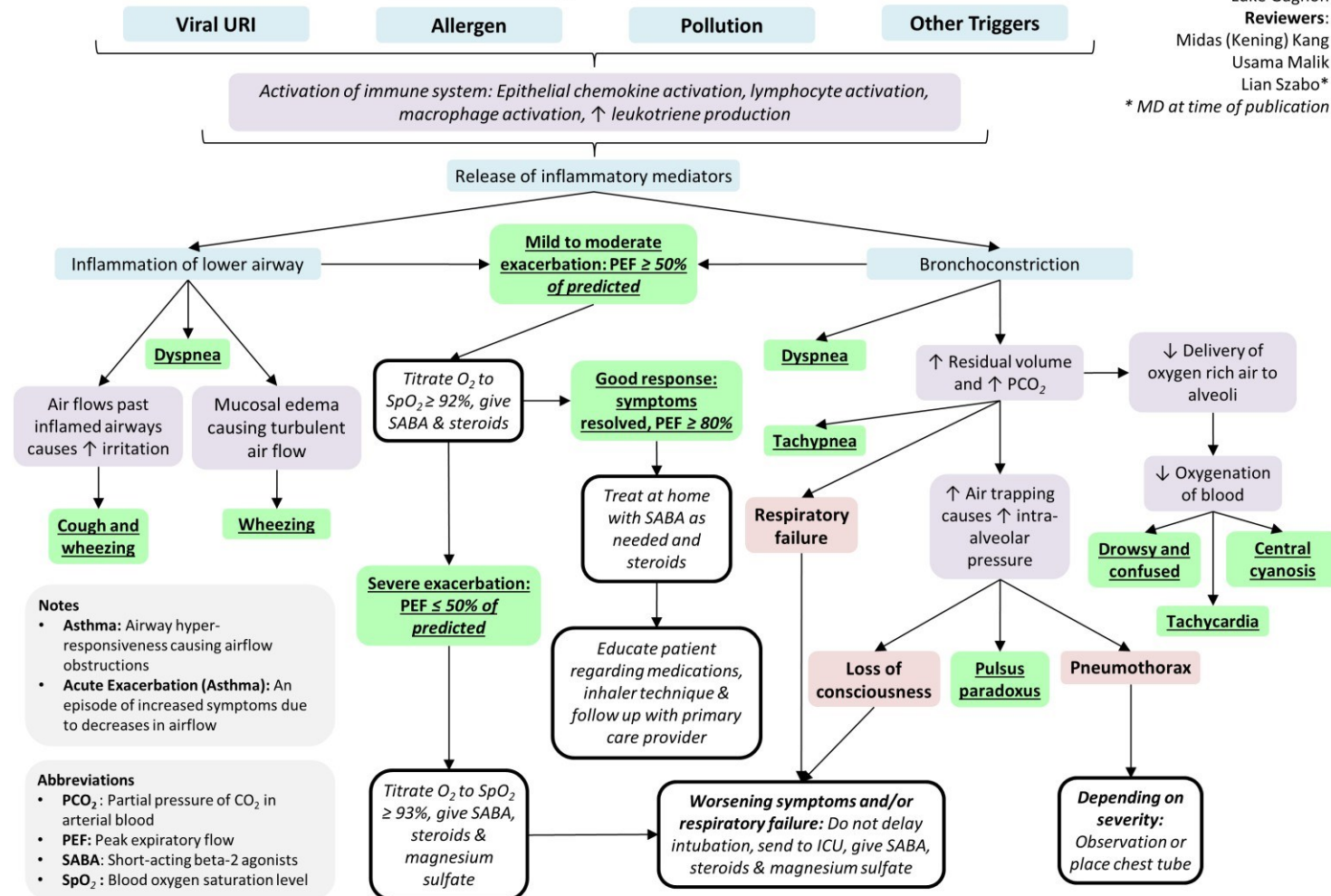
- **Asthma exacerbation:** a typically reversible episode of lower airway obstruction (bronchospasm) characterized by a worsening of asthma symptoms within a short period of time (acute or subacute) and accompanied by a change in baseline lung function
- **Bronchospasm:** a constriction in the bronchial muscles that results in airway obstruction within seconds to minutes. It is the characteristic symptom of asthma exacerbations, but it may also be triggered by certain medications or mechanical ventilation.
- **Status asthmaticus:** a term used to describe severe asthma exacerbations that progress rapidly and do not respond to standard therapy

Risk factors for asthma exacerbations

- Exposure to asthma triggers
- SABA-only treatment and/or overuse of SABA
- Inadequate inhaled corticosteroid (ICS) prescription and/or poor adherence to ICS
- Incorrect inhaler technique
- Prior intubation or ICU admission for asthma
- ≥ 1 severe exacerbations in the previous year
- $FEV_1 < 60\%$ of predicted average value
- High bronchodilator responsiveness
- Blood eosinophilia

Pathophysiology

Asthma Acute Exacerbation: Pathogenesis and Treatment



Clinical Features

- **Common symptoms**

- Dyspnea
- Increased coughing

- **Vital signs and general appearance**

- Tachypnea
- Tachycardia
- Pulsus paradoxus
- **Hypoxemia** ($\text{SpO}_2 < 90\%$ on ambient air, possible cyanosis)
- **Altered mental status**

- **Signs of bronchoconstriction**

- Prolonged expiratory phase
- Expiratory wheezing
- **Silent chest**
- Hyperresonance on percussion
- Inferior displacement and poor movement of the diaphragm

- **Signs of increased work of breathing:** e.g., use of accessory muscles

Management Acute Asthma Exacerbation

- **Severe to life-threatening asthma exacerbation**

- Urgent respiratory support

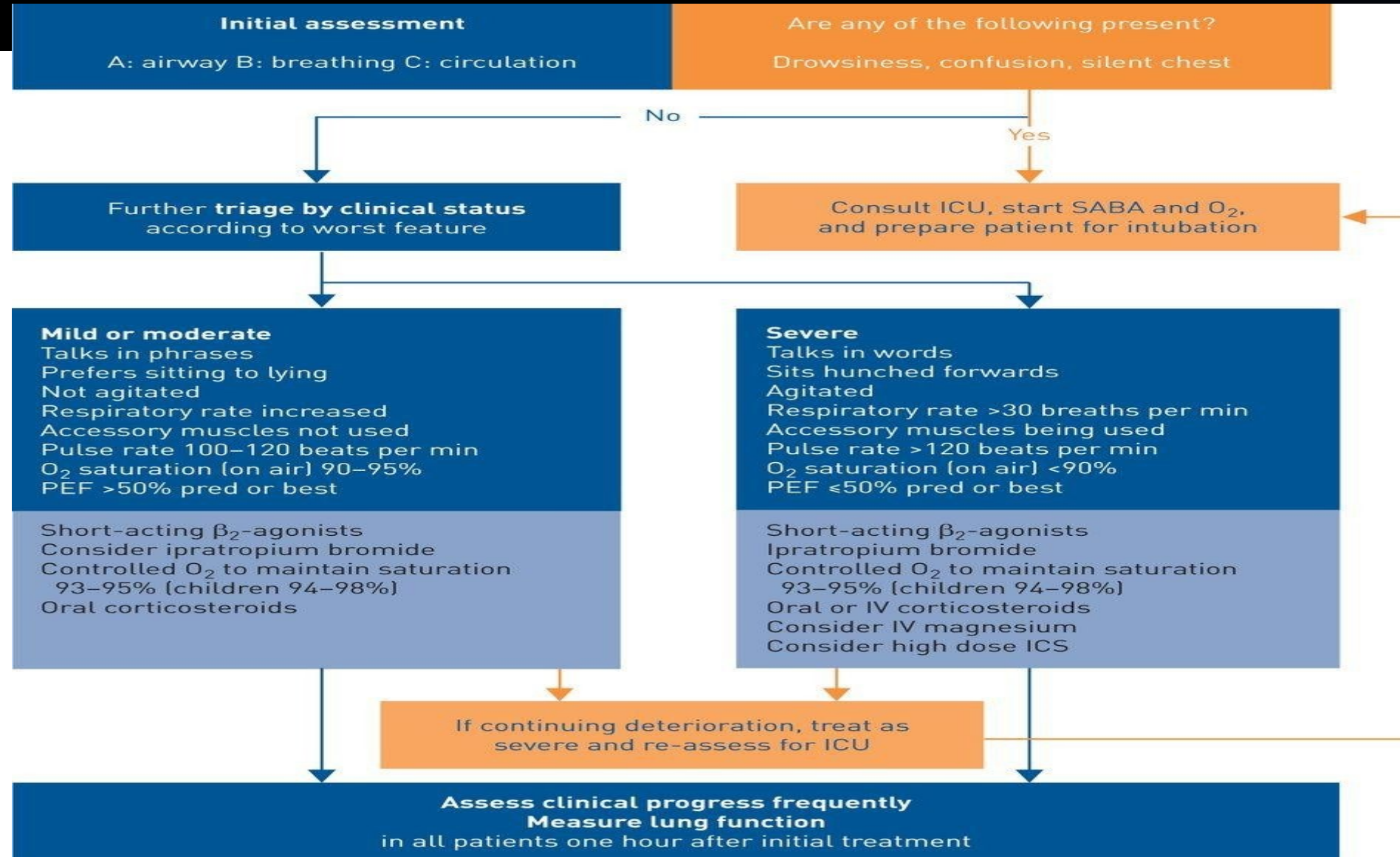
- Supplemental oxygen therapy
 - Prepare for intubation and mechanical ventilation in asthma
 - Monitor for signs of impending respiratory failure.

- **Maximize pharmacotherapy for acute asthma exacerbations.**

- Nebulized **SABA**
 - Plus nebulized **SAMA**
 - Plus oral or IV **glucocorticoids**
 - Consider a single dose of IV magnesium sulfate.
 - Consider ICS-containing therapy.

- **Consider fluid repletion (preferably oral) in infants and young children**

Acute Asthma Exacerbation



To Summarize

Figure 17.1 Asthma and COPD, summary of characteristics

Asthma

- Rarely smokes
- Symptoms evident usually < 35 years, often from childhood
- Variable breathlessness and PEFR from day to day
- Dry cough and wheeze, typically at night time



COPD

- Smoker or ex-smoker
- Symptoms usually evident > 50 years of age
- Persistent and progressive breathlessness
- Chronic productive cough, often worse in the morning

Spirometry

FEV₁/FVC ratio < 0.7 but a good response to bronchodilators returns this to normal

Spirometry

FEV₁/FVC ratio < 0.7 but does not respond to bronchodilators and remains at < 0.7

Spirometry: airflow obstruction is defined as reduced FEV₁/FVC ratio (< 0.7)
FEV₁ is the maximum amount of air forcibly exhaled in 1 second
FVC is the maximum amount of air that can be forcibly exhaled from maximal inspiration to expiration

Box 17.1 Factors that act as asthma triggers.

Source: RCP, 2014.

- Viral infections
- Medicines: aspirin, NSAIDS (i.e. ibuprofen, naproxen)
- Aero allergens: house dust mite, pollens, spores, animal dander
- Exercise
- Oral allergens: eggs, fish, milk, nuts
- Environmental factors: tobacco smoke, fumes, air pollutants, climatic variation

Box 17.2 Risk factors/causes of COPD

- Tobacco smoke
- Occupational pollutants, such as dust and chemicals
- Low socioeconomic status: factors, such as poor nutrition, and overcrowding
- Genetic factors: alpha-1 antitrypsin deficiency

Case Scenario 1

A 20-year-old man comes to the physician because of recurrent episodes of shortness of breath and a nonproductive cough for the past 4 months. He has two episodes per week, which resolve spontaneously with rest. Twice a month, he wakes up at night with shortness of breath. His pulse is 73/min, respirations are 13/min, and blood pressure is 122/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows no abnormalities. Spirometry shows an FVC of 95%, an FEV_1 :FVC ratio of 0.85, and an FEV_1 of 81% of predicted. Which of the following is the most appropriate initial pharmacotherapy?

- a. Albuterol inhaler
- b. Fluticasone inhaler
- c. Oral montelukast sodium
- d. Cromolyn sodium inhaler
- e. Mometasone inhaler and oral zafirlukast
- f. Salmeterol inhaler

Case Scenario 2

A 6-year-old boy is brought to the physician by his mother for coughing, nasal congestion, and intermittent wheezing for the past 2 months. The child has a history of eczema. Since birth, he has had three upper respiratory tract infections that resolved without treatment, and one episode of acute otitis media treated with antibiotics. His family moved into affordable housing 3 months ago. His temperature is 37.2°C (98.9°F), pulse is 120/min, respirations are 28/min, and blood pressure is 90/60 mm Hg. There are scattered wheezes on pulmonary examination. Which of the following is the most appropriate next step in management?

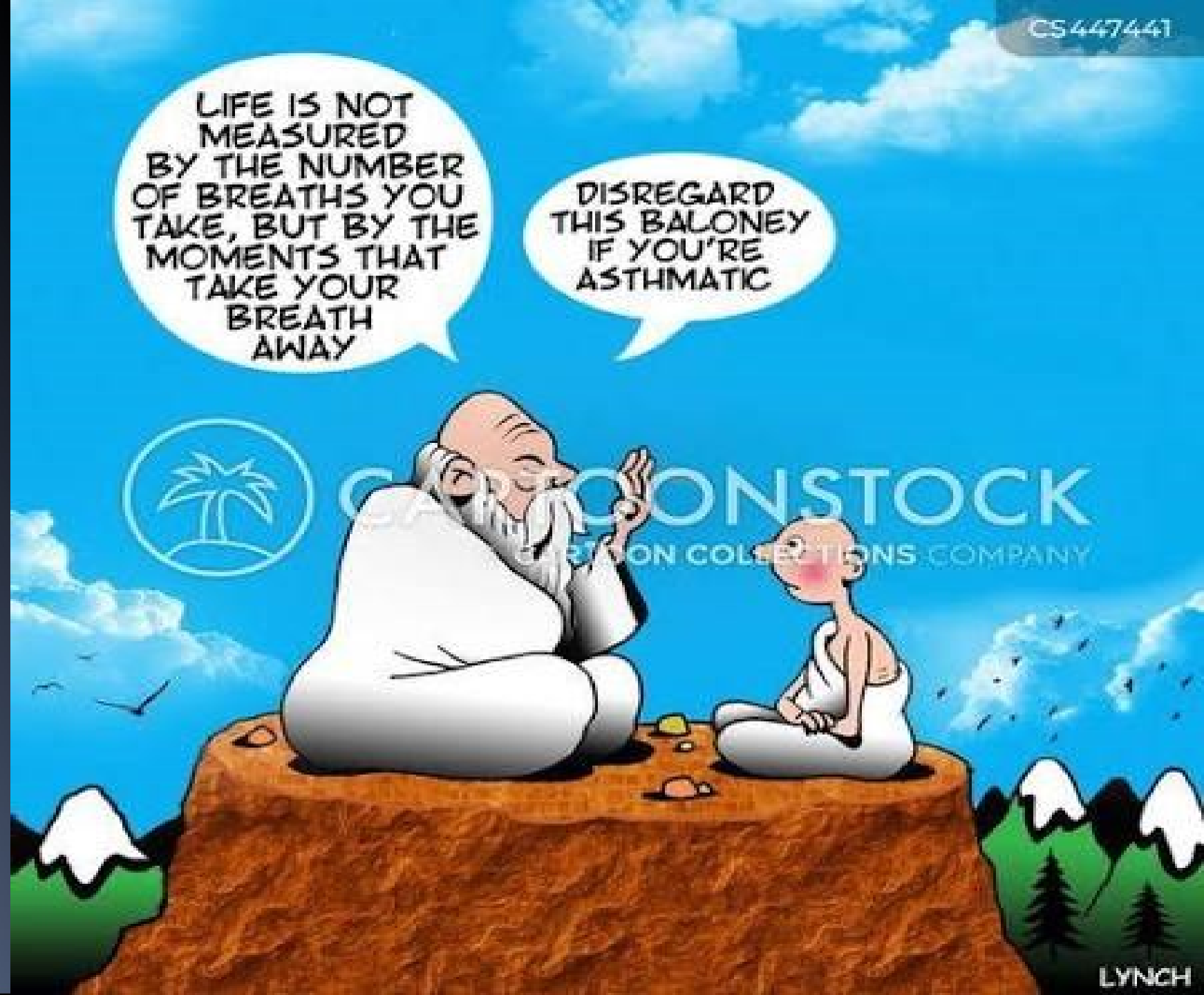
- a. Flow cytometry for B cells
- b. Spirometry
- c. Skin prick testing
- d. Throat culture
- e. Serum IgA levels
- f. Dihydrorhodamine 123 test

Case Scenario 3

A 6-year-old boy is brought to the emergency department by his mother for worsening wheezing and shortness of breath over the past day. He has not had a fever, cough, vomiting, or diarrhea. He has asthma and eczema. He uses a glucocorticoid inhaler and an albuterol inhaler but has missed his medications for the past week while on vacation. He appears uncomfortable. His temperature is 36°C (96.8°F), pulse is 120/min, respirations are 40/min, and blood pressure is 100/80 mm Hg. Expiratory and inspiratory wheezing is heard throughout both lung fields. There are moderate intercostal and subcostal retractions and a decreased inspiratory to expiratory ratio. Nebulized albuterol and ipratropium treatments and intravenous methylprednisolone are given in the emergency department for a presumed asthma exacerbation. One hour later, the child is limp and lethargic. Magnesium sulfate is administered. His temperature is 36°C (96.8°F), pulse is 150/min, respirations are 22/min, and blood pressure is 100/70 mm Hg. No wheezing is heard on repeat pulmonary examination. Which of the following is the most appropriate next step in management?

- a. Add leukotriene-receptor antagonists to nebulizer treatment
- b. Intubate with mechanical ventilation
- c. Add racemic epinephrine to nebulizer treatment
- d. Perform needle thoracostomy at the 2nd intercostal space
- e. Start noninvasive positive pressure ventilation
- f. Perform bronchoscopy
- g. Switch to nonracemic albuterol treatment
- h. Provide helium and oxygen mixture
- i. Provide additional dose of methylprednisolone

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board 4th edition

Thank You